

## Question # 1

 Revisit

Which of the following options is not a constituent of a Decision Tree?

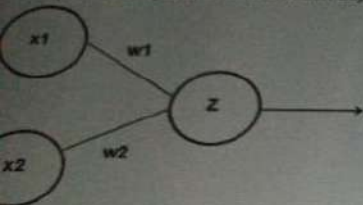
Choose the best option

- ☐ Internal Node
- ☐ Leaf Node
- ☐ Intermediate Gain Node
- ☐ Root Node

Question #3

Revisit

A Neural Network consists of an Input Layer, Hidden Layer and an Output Layer. Which of the given options forms the important components of the Hidden Layer?



Choose the best option

- ☐ Biases and Input neuron
- ☐ Weights and Output neurons
- ☐ Weights and Biases
- ☐ Weights and Input neuron

### Question # 2

How to attempt?

Question :

#### Coding Marathon

N number of people participated in a coding marathon where they were asked to solve some problems. Each problem carried 1 mark and at the end of the marathon, the total marks that each person achieved was calculated.

As an organizer, you have the list of the total marks that each person achieved. You have to calculate the sum of the marks of top K scorers from the list.

#### Input Specification:

**input1:** N, Total number of participants

**input2:** K, Top scorers

**input3:** An array of length N with the scores of all N participants

#### Output Specification:

Return S, sum of the marks of top K scorers from the list.

#### Example 1:

**input1:** 4

**input2:** 2

```
1
2 # Read only region start
3 class UserMainCode(object):
4     @classmethod
5     def marathon(cls, input1,
6
7         input1 : int
8         input2 : int
9         input3 : int[]
10
11     Expected return ty
12     ...
13     # Read only regio
14     # Write code her
15     pass
16
17
```

Revisit

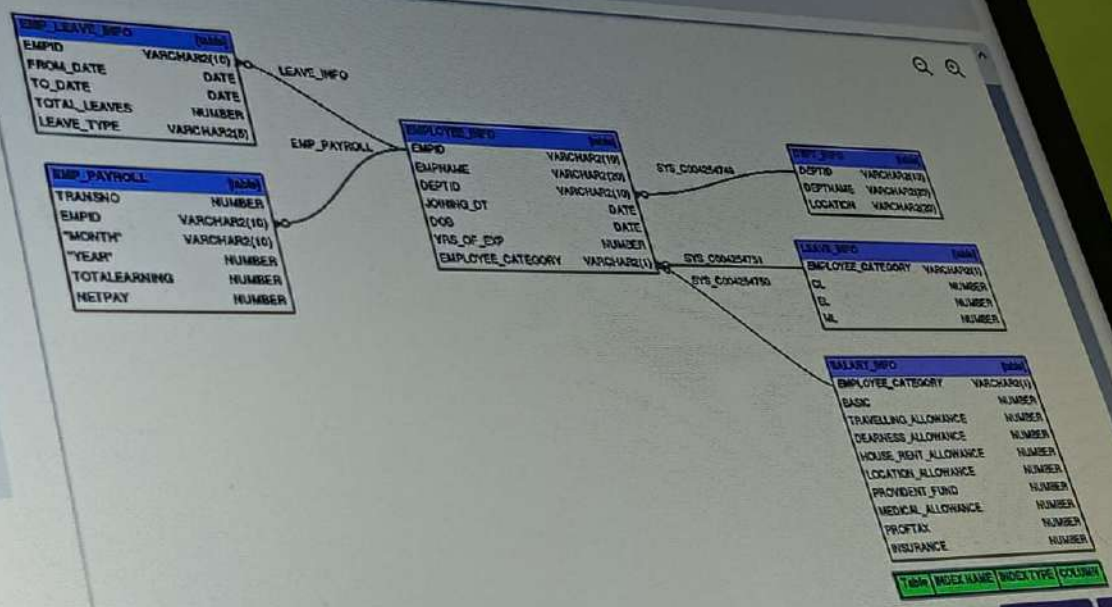
1 2 < 2 of 2 >

Total 00:56:39 Finish Test

Enter your response

each department (use  
ent ID.  
info" which contains three  
rgency  
mentioned order.

View Schema tab at the  
of the screen.



Query Output View Schema Test Cases

Run Query Check Query

Prev Question

### Question # 2

How to attempt questions:

Question:

Write a MySQL query to find:

The train ID and name of all trains that have a name starting with the alphabet 'M' and that go to the station with name "Pune".

Your output should contain 2 columns in the below-mentioned order.

| train_id | train_name |
|----------|------------|
|----------|------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.

Enter your response

MYSQL 5.7

1

Query Output View Schema



How to attempt?

Question :

Job

You just got a new job, but your new office has a weird rule. They allow you to take interval breaks in between tasks if there is no task available. But the problem is that the tasks come randomly and sometimes it may be required to do them simultaneously.

On your first day, you are given a list of tasks with their starting and ending time. Find out the total time you will get for breaks. Assume ending time to be greater than starting time.

Note:

The minimum start time and the maximum end time in the array is the total time duration he spent in the office.

Input Specification:

**input1:** Number of tasks

**input2:** 2-d array in form [t1, t2] representing starting and ending time period of the task

Output Specification:

```
1
2 # Read only region start
3 class UserMainCode(object):
4     @classmethod
5     def breakTime(cls, input1, in
6         ...
7         input1 : int
8         input2 : int[-]
9
10         Expected return type :
11         ...
12         # Read only region en
13         # Write code here
14         pass
15
16
```

How to attempt?

Question

Job

You just got a new job, but your new office has a weird rule. They allow you to take interval breaks in between tasks if there is no task available. But the problem is that the tasks come randomly and sometimes it may be required to do them simultaneously.

On your first day, you are given a list of tasks with their starting and ending time. Find out the total time you will get for breaks. Assume ending time to be greater than starting time.

**Note:**

The minimum start time and the maximum end time in the array is the total time duration he spent in the office.

**Input Specification:**

**input1:** Number of tasks

**input2:** 2-d array in form  $[t1, t2]$  representing starting and ending time period of the task

**Output Specification:**

Your function must return an integer representing the total break

```
1
2 # Read only region start
3 class UserMainCode(object):
4     @classmethod
5     def breakTime(cls, input1, input2):
6         ...
7         input1 : int
8         input2 : int[-]
9
10     Expected return type :
11     ...
12     # Read only region end
13     # Write code here
14     pass
15
16
```

## Question #7

Revisit

Choose the best option

Consider the given data. Which of the following regression model of  $x$  and  $y$  is used to calculate  $z$ ?

| X | Y  | Z  |
|---|----|----|
| 3 | 20 | 63 |
| 4 | 18 | 58 |
| 8 | 21 | 71 |
| 6 | 19 | 63 |
| 2 | 24 | 74 |

- ☐  $z = x + 2y$
- ☐  $z = x + 3y$
- ☐  $z = 10x + y$
- ☐  $z = x + y + 40$



Question # 10

Revisit

Choose the best option

If the parameters are indicated by the letter given, then which of the following formulas represent Accuracy?

True - T  
False - F  
Negative - N  
Positive - P

- ☐ Accuracy =  $\frac{TP+TN}{TP+FP+FN+TN}$
- ☐ Accuracy =  $\frac{TP+FN}{TP+FP+FN+TN}$
- ☐ Accuracy =  $\frac{FP+TN}{TN+FP+FN+TN}$
- ☐ Accuracy =  $\frac{FP+FN}{TP+FP+FN+TN}$

## Question # 4

Revisit

Choose the best option

If BIC of 4 different models is as follows, which model is the most preferable:

- Model 1:  $b$
- Model 2:  $2b$
- Model 3:  $b/2$
- Model 4:  $4b$

- ☐ Model 4
- ☐ Model 2
- ☐ Model 1
- ☐ Model 3

## Question # 5

Revisit

The given image is a dataset of one animal in each image. How will you label each animal image corresponding to the other image in Machine Language?



Choose the best option

- ☐ Categorical data
- ☐ Numeral vector
- ☐ Numeral vector and Categorical data
- ☐ Hot encoded vector

Question # 3

Revisit

Below are the two classes of dog image:



$P1 = [0.8, 0.15, 0.03, 0.02]$

$P2 = [0.87, 0.06, 0.03, 0.03]$

Choose the best option

- ☐ P2
- ☐ Both are equally close to perfect prediction
- ☐ Perfect prediction not possible for this class
- ☐ P1

Which class is closer to perfect prediction?

Question # 2

 Revisit

Choose the best option

If X buys 2 spoons and 1 bowl in 30 rupees; and 1 spoon and 2 bowls in 51 rupees, then what is the price of a spoon?

- ☐ 5
- ☐ 3
- ☐ 3.33
- ☐ 6



Question # 1

Revisit

If a feature has a highly skewed distribution, then which of the following transformations should be used to transform the data?

Choose the best option

- ☐ Standard Scaling
- ☐ Geometric Means Transformation
- ☐ Binning
- ☐ Log Transformations

## Question # 6

 Revisit

The user has a two-variable regression model where variable  $z$  is defined as a function of  $x$  and  $y$ . How will you represent this model geometrically?

Choose the best option

- ☐ A hyperbola
- ☐ A hypersphere
- ☐ A plane
- ☐ A straight line

## Question # 9

Revisit

Choose the best option

What will be mode value for the following dataset?

11, 12, 13, 15, 14, 16, 17, 18, 14, 19, 20

- ☐ 16
- ☐ 10
- ☐ 15
- ☐ 14

## Question # 8

Revisit

Choose the best option

A company produces  $X$  units of a product A and  $Y$  units of a product B making a profit of  $Z$ . Each unit of A and B is sold at Rs 15 and Rs 20 respectively. The company wants to maximize its profit.

To formulate a Linear optimization problem in this scenario, what will be the decision variables?

- ☐ X, Y, Z
- ☐ A, B, X, Y
- ☐ A, B, Z
- ☐ A, B

## Question # 11

Revisit

A company produces  $X$  units of a product A and  $Y$  units of a product B making a profit of  $Z$ . Each unit of A and B is sold at Rs 15 and Rs 20 respectively. The company wants to maximize its profit. To formulate a Linear optimization problem in this scenario, what will be the decision variables?

Choose the best option

- ☐ A, B, X, Y
- ☐ A, B, Z
- ☐ X, Y, Z
- ☐ A, B

MYSQL 5.7

```
1 select train_id,train_name from train_details_tbl where train_name like "M%" and train_to in  
2 (select distinct station_id from train_stations_tbl where station_name = "Pune");
```



```
12 SELECT DEPTID, avg(leave) as Avg_Leave
13 from
14 (
15     SELECT DEPTID, CL+EL+ML as leave from EMPLOYEE_INFO, LEAVE_INFO
16     where EMPLOYEE_INFO.EMPLOYEE_CATEGORY=LEAVE_INFO.EMPLOYEE_CATEGORY
17 )
18 group by(DEPTID);
19
```

Query Output View Schema Test Cases

Run